Vibration-Based Bearing Fault Detection for an Industrial Centrifuge

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1 Introduction and Background

The machine in this study is a disc-stack centrifugal separator driven by an electric motor. Its bowl spins at high speed to separate liquids and solids by density. Centrifuges combine heavy rotors, high speed and a load that changes as material is fed in and discharged, so small faults grow quickly.

1.1 Vibration analysis and bearing faults

Vibration is the most informative signal for monitoring rotating machines, because most mechanical faults change how a machine vibrates long before they cause a failure.
Each fault leaves its own pattern in the vibration signal:

- **Imbalance and misalignment** appear at the running speed (1X) and its low multiples.
- **Mechanical looseness** produces many harmonics of running speed.
- **Rolling-bearing defects** produce a short impact each time a rolling element passes the damage. These impacts are weak but excite structural resonances at kilohertz frequencies, and they repeat at characteristic defect frequencies set by the bearing geometry.

Because bearing impacts are weak and buried under rotor vibration, the standard detection method is **envelope analysis** [10]. The signal is band-pass filtered around a resonance the impacts excite, its envelope is extracted, and the envelope's spectrum reveals the impact repetition rate.

1.2 Machine learning for fault detection

Machine learning automates the step from vibration features to a decision. Reviews describe a pipeline of feature extraction, feature selection and a classifier or anomaly detector. Deep learning can learn features directly from raw signals, but systematic reviews note that near-perfect accuracies on laboratory test rigs often fail to carry over to real critical machinery, mainly because labelled fault data is scarce and models are hard to interpret. Anomaly detection, which learns only what normal looks like, suits industrial settings where healthy data is plentiful and failures are rare.

1.3 Aims and structure

This report answers the three questions in the following sections:

1. Exploratory analysis (Q1): understand the data, find patterns and anomalies, and form hypotheses about the labelled fault.
2. Modelling (Q2): build and evaluate a model that detects signs of bearing fault.
3. Architecture (Q3): design a production system.

2 Exploratory data analysis (Q1)

2.1 Data and methods

The dataset used in this analysis contains 26,502 rows of motor drive-end (DE) sensor records from a centrifuge line. Most columns are sparse, each row usually carries only one kind of reading. The brief labels 1 to 4 January 2025 as a bearing fault.

- **Raw vibration waveforms.** Motor DE_Z Short Wave High Frequency (16,384 samples), Motor DE_Z Long Wave High Frequency (131,072), Motor DE_Z Short Wave Low Frequency (9,216), and Motor DE_X Short Wave Low Frequency and Motor DE_Y Short Wave Low Frequency (5,120 each). Each is present in about 10% of rows, except the long-wave channel at about 2%.
- **Summary vibration values.** Motor DE_Z Short Wave High Frequency RMS is the best-covered column (73% of rows). Motor DE_Z Short Wave Low Frequency RMS appears in 13% of rows, and Motor DE_X Short Wave Low Frequency RMS and Motor DE_Y Short Wave Low Frequency RMS in under 1%.
- **Speed.** Motor DE_Z Speed Total Value (3% of rows), Motor DE_X Speed Total Value and Motor DE_Y Speed Total Value (under 1%).
- **Temperature.** Motor DE_Temperature, a single reading per record, in about 10% of rows.
- **Audio and magnetic.** Motor DE_Audio Signal (131,072 samples, about 2% of rows), and Motor DE_Audio Decibels and Motor DE_Magnetic Leakage RMS (5 rows each).
- **Timestamp.** date, present in every row.

Note: *The assessment document specifies that data from 29 April to 9 June 2025 is reserved for evaluation and indicates that this period would not be provided. However, the data for this period was present in the supplied file but was excluded from the analysis.*

2.2 Findings

Vibration RMS separates idle from running

The Z-axis high-frequency RMS in the development set (n = 11,372) has two clear groups: one near 0 and one around 20–35. K-means placed the cluster centers at 0.19 and 26.54, giving a midpoint cutoff of 13.37. Few readings fall between 5 and 15, so the exact cutoff matters little. Overall RMS is widely used as a simple condition indicator in vibration monitoring [3], which supports using it to mark operating state.

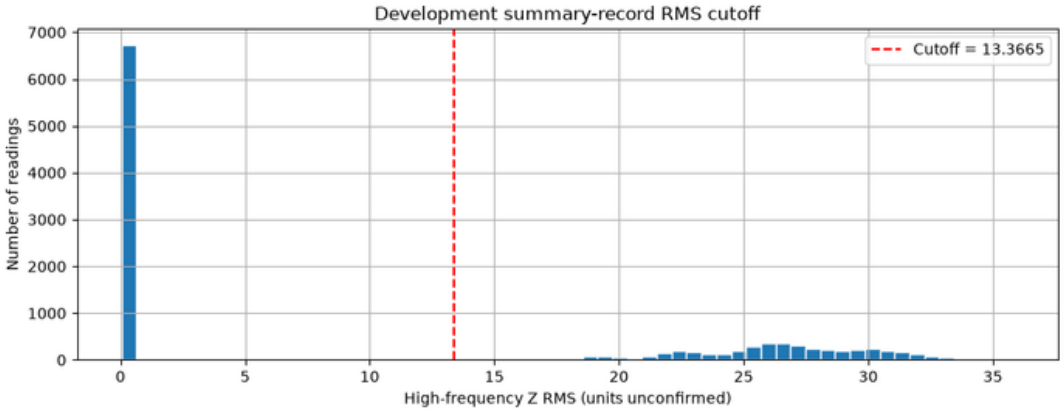


Figure 1. Distribution of Z-axis high-frequency RMS in the development set.

The fault run and the last six runs share high RMS levels

The development set holds 21 running periods between 2 January and 28 April 2025. The labelled fault run (run 1) has a median RMS of 29.7. Runs 16–21 (24 March – 28 April) have medians of 28.2–31.6, while runs 2–15 range from 13.2 to 26.8. This raises the possibility that runs 16–21 may exhibit a fault, can categorized as suspicious. However, an increase in overall RMS is not specific to bearing damage, therefore, this observation is be treated as a hypothesis for further investigation rather than a definitive diagnosis. [2].

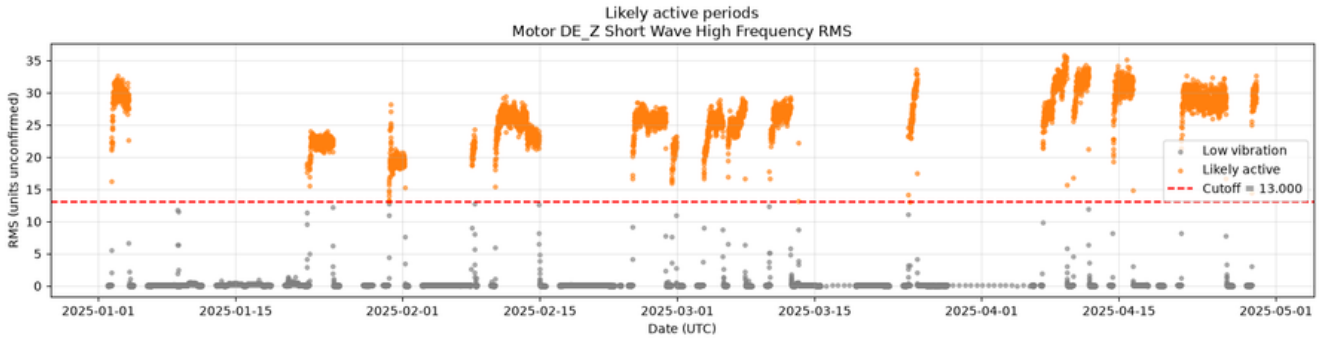


Figure 2. Z-axis high-frequency RMS from January to April 2025

Temperature alone is likely to give false alarms

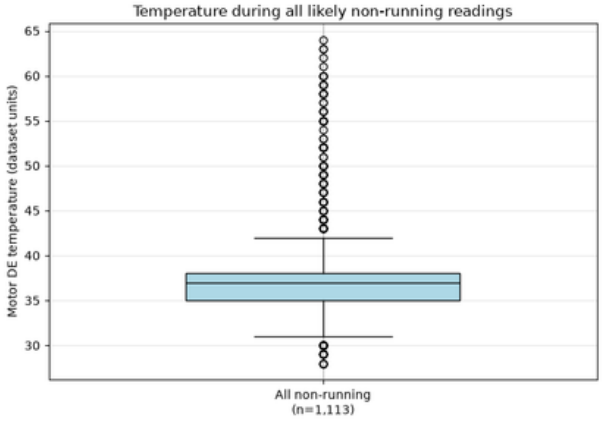


Figure 3. Motor drive-end temperature distribution

Of 1,984 temperature readings, 1,113 were matched to idle-level RMS (below 13). Their median was 37 (interquartile range 35–38), but some reached 63. High readings while the motor was likely stopped mean a fixed temperature limit could trip without any fault. A review of temperature-based monitoring finds it useful but limited, and best read alongside vibration [5]. Whether temperature helps here needs a comparison of healthy and fault periods under similar running conditions.

Fault-period differences sit in specific frequency bands

Raw waveforms (figure 4) already showed a visible difference on the X-axis low-frequency channel between the fault and healthy periods. Band-level spectra then pointed to five candidate features. Values below are approximate group medians read from the notebook's plots.

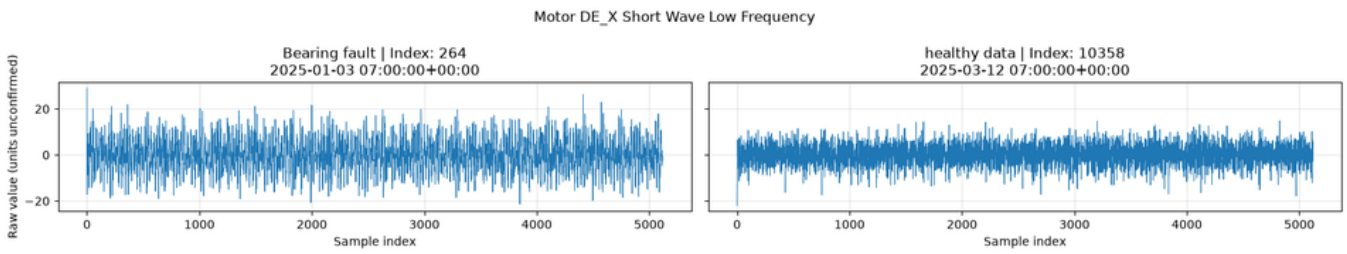


Figure 4. Raw X-axis low-frequency waveform (5,120 samples) for one fault-period recording (3 January 2025, 07:00 UTC, left) and one healthy recording (12 March 2025, 07:00 UTC, right).

Hypothesis 1 (H1): The labelled fault period has a distinctive X-axis concentration at 100–500 Hz

The X-axis 100–500 Hz band has a median acceleration band RMS of approximately 0.60 during the fault-labelled period, compared with 0.27 during healthy operation. Its median power fraction is also substantially higher: approximately 82% versus 45%. This suggests that the difference involves both increased vibration magnitude and a concentration of spectral power within this band. Therefore, `x_acc_100_500` is a candidate feature for distinguishing the labelled fault period from healthy operation. The suspicious period does not reproduce the same high power fraction, suggesting that its vibration pattern may differ from the known-fault pattern.

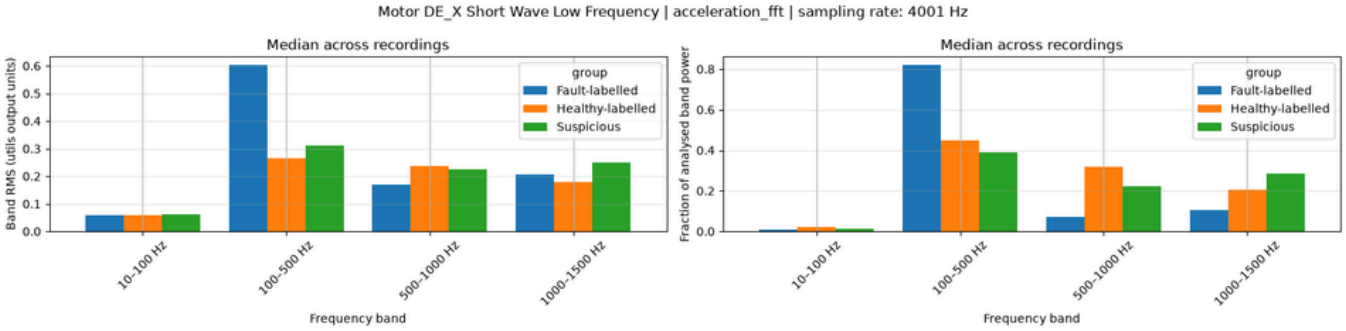


Figure 6. Acceleration spectrum of the X-axis low-frequency channel (sampled at 4,001 Hz). Median band RMS (left) and share of analysed power (right) by group.

Hypothesis 2 (H2): Increased Z-axis power at 2500–5000 Hz indicates a departure from healthy operation

The Z-axis short-wave 2500–5000 Hz band has a median band RMS of approximately 2.2–2.3 in the fault-labelled and suspicious periods, compared with approximately 1.45 in the healthy period. Both groups also show a higher power fraction than the healthy group. A similar pattern appears in the Z-axis long-wave recordings, providing consistency across recording lengths. Therefore, `z_acc_2500_5000` is a candidate feature for identifying departures from the healthy baseline. The suspicious period's resemblance to the fault-labelled period in this band motivates further investigation, but does not establish that both periods contain the same fault.

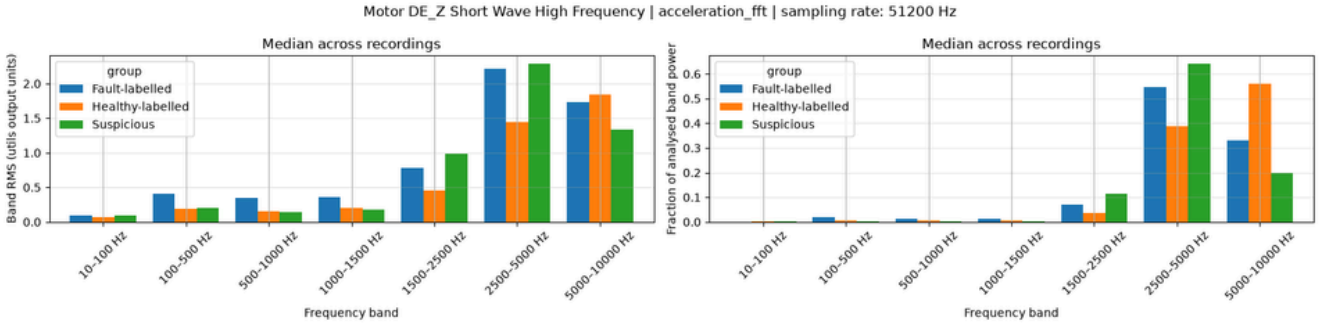


Figure 5. Acceleration spectrum of the Z-axis short-wave high-frequency channel (sampled at 51,200 Hz). Median band RMS (left) and share of analysed power (right) for fault-labelled, healthy-labelled and suspicious recordings.

Hypothesis 3 (H3): Y-axis 100–500 Hz provides supporting evidence of a spectral shift

The Y-axis 100–500 Hz median band RMS increases from approximately 0.28 in the healthy period to 0.35 in the suspicious period and 0.46 in the fault-labelled period. Its median power fraction is approximately 40%, 65%, and 71%, respectively. This suggests that the suspicious period shares some characteristics of the fault-labelled period on the Y axis. Therefore, `y_acc_100_500` is a supporting candidate feature. Its additional value should be assessed because X and Y features may contain overlapping information.

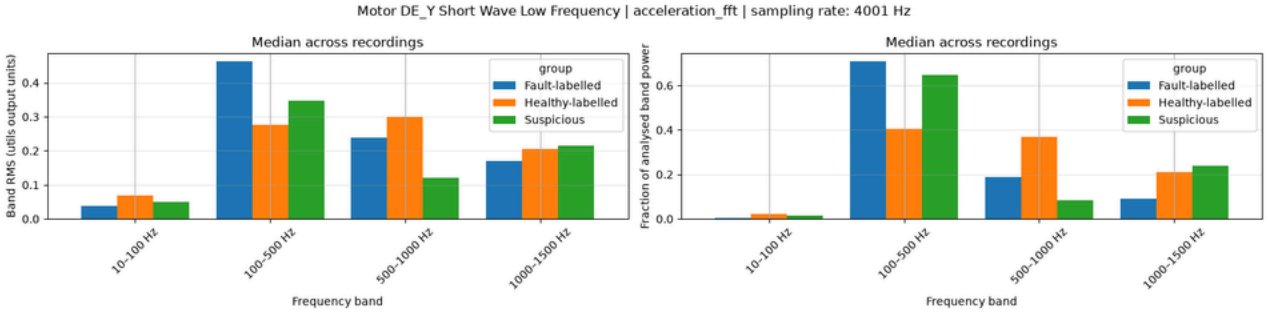


Figure 7. Velocity spectrum of the Y-axis low-frequency channel (sampled at 4,001 Hz). Median band RMS (left) and share of analyzed power (right) by group.

Hypothesis 4 (H4): X-axis velocity at 100–500 Hz distinguishes the labelled fault period from healthy operation

The X-axis 100–500 Hz median band RMS is approximately 5.7 during the fault-labelled period, compared with 1.7 during healthy operation and 2.6 during the suspicious period. The corresponding median power fractions are approximately 87%, 33%, and 55%, respectively. Meanwhile, the 10–100 Hz median band RMS remains similar across the three groups. This suggests that the fault-labeled difference is concentrated in 100–500 Hz rather than representing a uniform increase across all frequency bands. Therefore, `x_vel_100_500` is a strong candidate feature for distinguishing the labelled fault period from healthy operation.

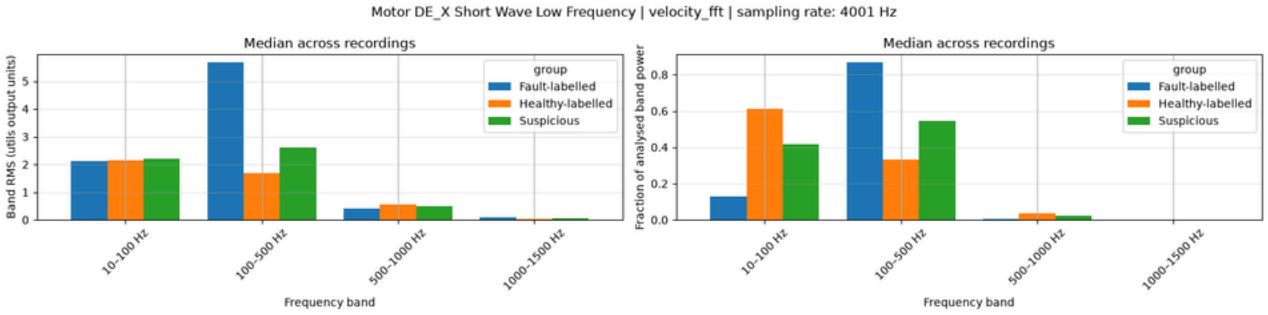


Figure 8. Velocity spectrum of the X-axis low-frequency channel (sampled at 4,001 Hz). Median band RMS (left) and share of analysed power (right) by group.

Hypothesis 5 (H5) : Y-axis velocity shows a redistribution of spectral power toward 100–500 Hz

The Y-axis 100–500 Hz median band RMS is approximately 3.1 during the fault-labelled period, compared with 1.6 during healthy operation and 2.6 during the suspicious period. Its median power fraction increases from approximately 20% in the healthy group to 54% in the suspicious group and 83% in the fault-labeled group. Conversely, the 10–100 Hz median band RMS is lower in the fault-labelled group than in the healthy group. This suggests that the labelled fault period involves a change in spectral distribution, alongside increased vibration within the selected band. Therefore, `y_vel_100_500` is a complementary candidate feature, particularly when expressed as a power fraction

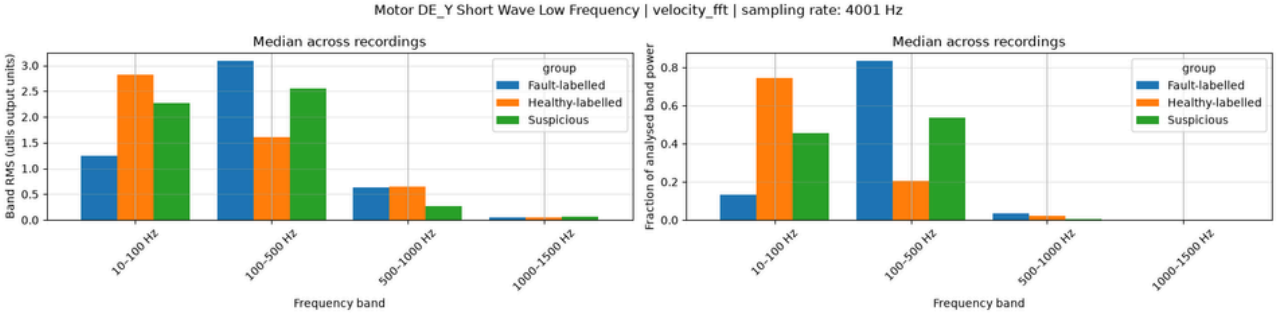


Figure 8. Velocity spectrum of the Y-axis low-frequency channel (sampled at 4,001 Hz). Median band RMS (left) and share of analysed power (right) by group.

3 Model Training (Q2)

3.1 Feature engineering and selection

The five frequency-band hypotheses in Section 2 were converted into numerical features using the provided feature_utils.get_features function and additional band-summary and envelope utilities. Acceleration processing converts the waveform to g, applies a 10 Hz high-pass filter, detrends the signal, trims the initial settling interval and removes its mean. Velocity processing converts acceleration to mm/s², integrates it, applies a 10–1000 Hz bandpass, detrends and trims the signal. These conversions assume that the original acceleration units are m/s². Sampling rates were 51,200 Hz for Z high-frequency, 5,120 Hz for Z low-frequency and 4,001 Hz for X and Y low-frequency waveforms.

The utilities produce processed RMS values and Welch spectra expressed as RMS amplitude per frequency bin. For each hypothesised band, band RMS is the square root of the summed squared bin amplitudes, while power fraction is the band's summed power divided by power in a specified reference range. The acceleration and velocity features use 10–1500 Hz as the reference range for X and Y; Z high-frequency acceleration uses 10–10000 Hz. Consequently, the Z power fraction was recalculated rather than copied from the broader normalization used in the Q1 chart. Velocity fractions describe the processed spectrum, whose power above 1000 Hz is attenuated by filtering.

An additional feature was engineered from the Z short-wave high-frequency signal. After acceleration preprocessing, a zero-phase 1800–4500 Hz bandpass selects a carrier band, and the magnitude of the Hilbert transform extracts its envelope. After trimming 5% from each boundary and removing the mean, a Hann/Welch spectrum is calculated. Its median RMS spectral-bin amplitude between 5 and 500 Hz summarizes modulation in the high-frequency signal. This extends the ordinary band features; it is neither integrated envelope-band RMS nor a measurement of a specific bearing defect frequency.

From these features were made

- X acceleration 100–500 Hz RMS. ***(H1)***
- X acceleration 100–500 Hz power fraction. ***(H1)***
- Z acceleration 2500–5000 Hz RMS ***(H2)***
- Z acceleration 2500–5000 Hz power fraction ***(H2)***
- Y acceleration 100–500 Hz power fraction ***(H3)***
- X velocity 100–500 Hz power fraction ***(H4)***
- Y velocity 100–500 Hz power fraction ***(H5)***
- Z HF envelope median bin amplitude, 5–500 Hz

A baseline Isolation Forest was trained using all eight features. An effect-size table was then used to compare each feature’s values in the fault-labelled and suspicious groups against healthy training recordings. The Fault–Healthy and Suspicious–Healthy columns show standardized mean dif ferences: the difference between group means divided by their combined variability. Positive values indicate higher values than healthy operation; negative values indicate lower values. Larger absolute values indicate greater separation between groups. The Weaker Effect Magnitude is the smaller absolute value of the two comparisons.

It highlights features that differ substantially from healthy operation in both groups, rather than only during the known fault episode.

Feature	Fault-Healthy	Suspicious-Healthy	Weaker Effect magnitude
X acceleration 100–500 Hz RMS	4.91	0.62	0.62
X acceleration 100–500 Hz power fraction	2.97	-0.06	0.06
Z acceleration 2500–5000 Hz RMS	5.01	4.08	4.08
Z acceleration 2500–5000 Hz power fraction	3.09	2.58	2.56
Y acceleration 100–500 Hz power fraction	2.07	1.57	1.57
X velocity 100–500 Hz power fraction	4.42	1.22	1.22
Y velocity 100–500 Hz power fraction	7.72	2.32	2.32
Z HF envelope median bin amplitude, 5–500 Hz	3.60	3.30	3.30

The four-feature configuration retained the four baseline features with the largest weaker-effect magnitudes: Z band RMS (4.08), the envelope feature (3.30), Z band power fraction (2.58), and Y velocity power fraction (2.32). All increased in both comparison groups. Together, they describe high-frequency vibration magnitude, its spectral distribution, envelope modulation and low-frequency velocity redistribution. This provides a rationale for retaining H2 and H5 alongside the envelope feature.

I removed the remaining four baseline features because their changes were less consistent or smaller across the two groups. X acceleration power fraction increased during the January fault but changed almost negligibly in the opposite direction during the suspicious period (2.97 versus −0.06). X acceleration band RMS also showed a much smaller suspicious effect (0.62) than its fault effect (4.91). X velocity power fraction and Y acceleration power fraction had weaker-effect magnitudes of 1.22 and 1.57, below those of the retained features. Their original hypotheses remain plausible, but they were less compelling for this reduced anomaly-screening configuration.

3.2 Model selection and justification

An Isolation Forest, an anomaly detection technique was selected because healthy recordings provide a reference distribution, while confirmed fault examples are confined to one short episode. Fitting a supervised fault classifier to that episode could learn characteristics specific to its operating conditions. Isolation Forest instead learns the healthy feature distribution and assigns higher anomaly scores to unusual combinations of features. Its tree-based splits do not require feature scaling.

3.3 Training procedure

Records from **29 April through 9 June 2025** were reserved as holdout data. Development recordings were grouped into the supplied January fault, healthy and suspicious periods. Only recordings with finite calculated Z short-wave high-frequency RMS **greater than 13** were selected as likely running observations. Features were joined using original row identifiers and timestamps; missing or non-finite required features were excluded rather than imputed.

For a controlled feature comparison, recordings had to be complete across the union of the three configurations' inputs. This retained 428 of 430 healthy recordings, all 39 fault-labelled recordings, and 271 of 276 suspicious recordings. Operating runs were inferred from calculated RMS **at least 13**, with a maximum gap of four hours. Healthy runs were ordered chronologically, and the latest approximately 25% were reserved for threshold calibration. Keeping entire runs together reduces leakage between neighboring recordings within a run, although the inferred run boundaries remain an approximation.

Item	Configuration
Healthy training population	335 recordings; run IDs 2–9
Healthy calibration population	93 recordings; run IDs 10–12
Fault-labelled development evaluation	39 recordings; one January episode
Suspicious development assessment	271 recordings; excluded from labelled metrics
Isolation Forest parameters	300 trees; `max_samples="auto"`; `contamination="auto"`; random seed 42
Model fitting	Healthy training recordings only; no feature scaling
Anomaly score	Negative of `score_samples`; larger values are more anomalous
Threshold calibration	95th percentile of healthy calibration scores; nominal 5% false-alarm target
Selected-model decision	Flag when anomaly score > **0.620584**

3.4 Evaluation results and interpretation

Labelled development evaluation combined 93 healthy calibration recordings as negatives and 39 fault-labelled recordings as positives. Suspicious recordings were assessed separately.

- Threshold: 0.620584
- Healthy false-alarm rate: 5/93 (5.38%)
- Fault-labeled recall: 100%
- Fault-labeled precision: 88.64%
- F1 score: 93.98%
- Accuracy: 96.21%
- Balanced accuracy: 97.31%
- ROC AUC : 1

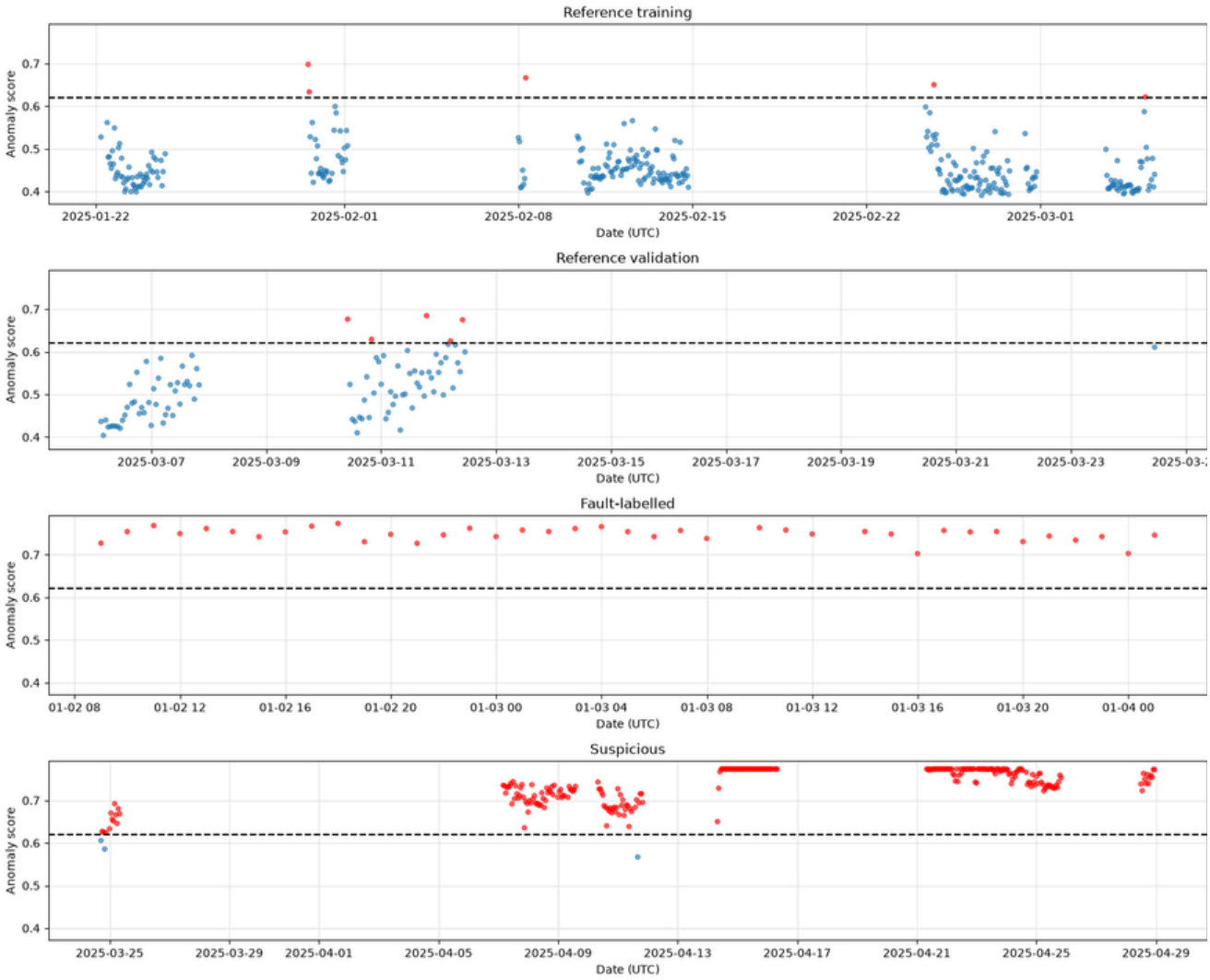


Figure 9. Predictions according to the selected threshold.

4 System Architecture (Q3)

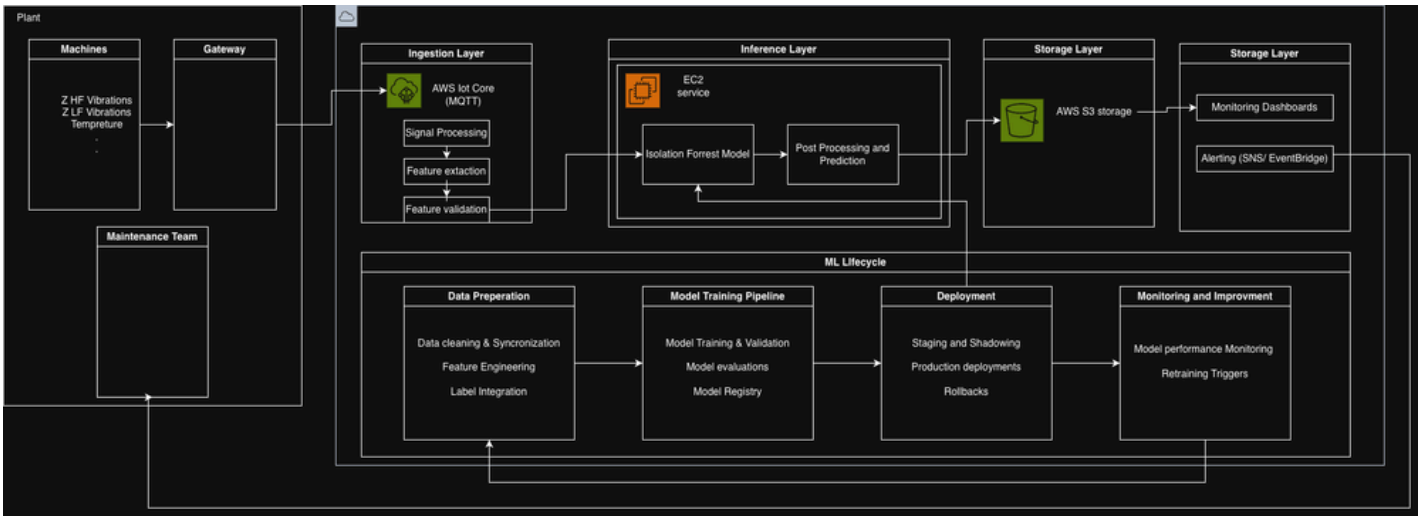


Figure 10. Proposed Architecture

4.1 Data Ingestion and Storage

The centrifuge machines generate multiple sensor readings, including high-frequency and low-frequency vibration measurements and temperature data. These measurements are collected by a gateway located within the plant.
The gateway forwards the sensor data to AWS IoT Core using MQTT, which serves as the cloud ingestion point.

After ingestion, the sensor data passes through three processing stages:

1. Signal Processing: Processes the received vibration signals to prepare them for analysis.
2. Feature Extraction: Extracts the numerical vibration characteristics required by the Isolation Forest model.
3. Feature Validation: Checks the extracted features before passing them to the inference layer.

The validated features are then forwarded to the Isolation Forest model hosted on an Amazon EC2 service. After inference and post-processing, the results are stored in Amazon S3. The stored information is subsequently used by the monitoring dashboard.
The diagram shows S3 storage for inference outputs. It does not explicitly show a separate raw-waveform storage flow or database for processed sensor records.

4.2 ML Life Cycle

Data Preparation: This component performs data cleaning and synchronization, feature engineering, and label integration. It prepares historical sensor data and newly available labelled information for model training.
Model Training Pipeline: The prepared data is passed to the training pipeline, which performs model training and validation, model evaluation, and model registration. The model registry manages the different versions of the trained model.
Deployment: The deployment component includes staging and shadowing, production deployment, and rollback capabilities. Models are evaluated before being deployed to the production inference environment, and previous versions can be restored if necessary.
Monitoring and Improvement: This component monitors model performance and identifies retraining triggers. Its output feeds back into data preparation, forming a continuous improvement cycle.

4.3 Inference Strategy

The proposed architecture uses an event-driven near-real-time inference strategy. Once the data is received through AWS IoT Core, it undergoes signal processing, feature extraction, and validation before being forwarded to the Isolation Forest model hosted on Amazon EC2.

The model generates an anomaly score as soon as a complete, temporally aligned feature vector becomes available. Since the selected high-frequency vibration channel is uploaded every four hours, predictions are expected approximately every four hours, subject to the availability of the other required sensor channels.

Near-real-time inference is selected over continuous real-time processing because the sensor data is acquired and uploaded periodically rather than streamed continuously. Batch inference is not preferred because it would introduce additional delays between receiving sensor data and identifying abnormal vibration patterns.

4.4 Alerting and Maintenance Integration

The architecture contains a monitoring dashboard and an alerting service based on Amazon SNS and EventBridge.
The monitoring dashboard receives information from the Amazon S3 storage layer, allowing the results of model inference to be presented for monitoring purposes.

The alerting component represents the mechanism used to communicate significant events or anomalous machine conditions. SNS and EventBridge are intended to support event-driven notifications.
The alerting service is connected to the maintenance team at the plant. This enables maintenance personnel to receive notifications and investigate machines exhibiting abnormal vibration behavior.

4.5 Scalability

The architecture separates sensor acquisition, ingestion, inference, storage, and alerting to support independent scaling.

At the initial deployment of 50 machines, lightweight feature records produce relatively low cloud inference demand. The Isolation Forest is computationally inexpensive because predictions operate on four numerical features rather than full vibration waveforms.

For 500+ machines, edge processing capacity can be expanded by adding industrial gateways, while AWS IoT Core handles additional connected devices. Multiple inference workers can be deployed behind a load balancer, with an SQS queue introduced between ingestion and inference to buffer bursts and support reliable processing.

Amazon S3 provides scalable storage for historical recordings and model artifacts, while a time-series database supports fleet-level monitoring. Gateway buffering and retry mechanisms protect against temporary network disruptions. Machine identifiers, timestamps, event IDs, and model versions support traceability and duplicate detection.

This architecture balances bandwidth efficiency, operational reliability, centralized model management, and scalability. Its modular design allows additional sensor channels, machine types, and improved models to be introduced without replacing the overall system.

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